ClipSAT : Multivariable Calculus Partials, Multiple Integrals & Vector Calcu [ANSWER KEY]

Unit 4.01: Double Integrals, Polar Coordinates & Jacobians

Q1. Compute iint_R dA for R=[0,4]times[0,3] the area of R .

[x] A: 12
[ ] B: 7
[ ] C: 14
[ ] D: 6

Solution: Applying Double Integrals principles, text{Area}=4times3=12 . Therefore, the verified correct option is 12.

Q2. Compute iint_R dA for R=[0,2]times[0,4] the area of R .

[x] A: 8
[ ] B: 6
[ ] C: 12
[ ] D: 4

Solution: Applying Double Integrals principles, text{Area}=2times4=8 . Therefore, the verified correct option is 8.

Q3. Compute iint_R dA for R=[0,4]times[0,5] the area of R .

[x] A: 20
[ ] B: 9
[ ] C: 18
[ ] D: 10

Solution: Applying Double Integrals principles, text{Area}=4times5=20 . Therefore, the verified correct option is 20.

Q4. Compute iint_R dA for R=[0,2]times[0,8] the area of R .

[ ] A: 10
[x] B: 16
[ ] C: 20
[ ] D: 8

Solution: Applying Double Integrals principles, text{Area}=2times8=16 . Therefore, the verified correct option is 16.

Q5. Compute iint_R dA for R=[0,7]times[0,8] the area of R .

[ ] A: 15
[ ] B: 30
[x] C: 56
[ ] D: 28

Solution: Applying Double Integrals principles, text{Area}=7times8=56 . Therefore, the verified correct option is 56.

Q6. Compute iint_R dA for R=[0,8]times[0,3] the area of R .

[ ] A: 11
[ ] B: 22
[ ] C: 12
[x] D: 24

Solution: Applying Double Integrals principles, text{Area}=8times3=24 . Therefore, the verified correct option is 24.

Unit 4.01: Double Integrals, Polar Coordinates & Jacobians

Q7. Compute iint_R dA for R=[0,1]times[0,3] the area of R .

[ ] A: 4
[ ] B: 8
[x] C: 3
[ ] D: 1

Solution: Applying Double Integrals principles, text{Area}=1times3=3 . Therefore, the verified correct option is 3.

Q8. Compute iint_R dA for R=[0,5]times[0,3] the area of R .

[ ] A: 8
[ ] B: 16
[x] C: 15
[ ] D: 13

Solution: Applying Double Integrals principles, text{Area}=5times3=15 . Therefore, the verified correct option is 15.